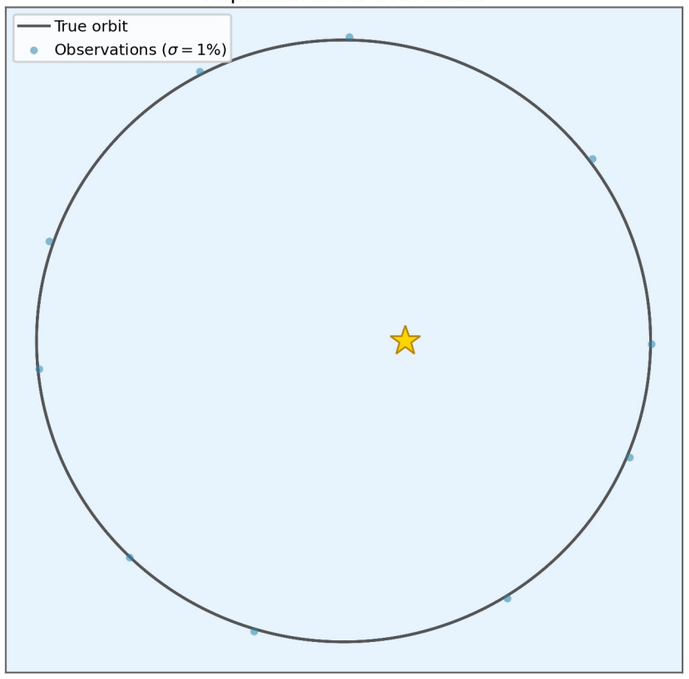
MinAction.Net: Discovering Force Laws from Noisy Orbits

INPUT

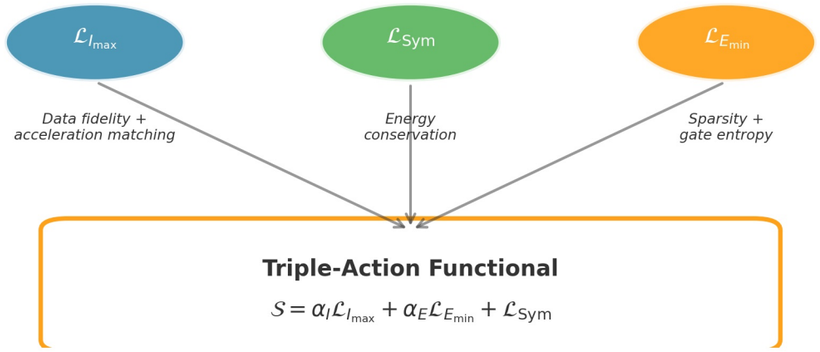
Noisy observations

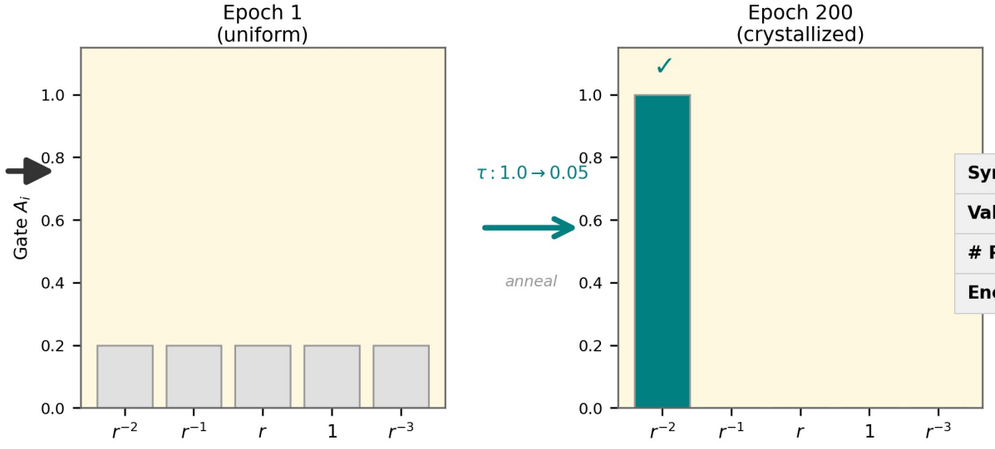
Keplerian orbit with noise



METHOD

Triple-Action Learning





OUTPUT

Discovered physics

Kepler

$F = -\frac{GM}{r^2}\hat{r}$

$\hat{\theta} = 0.94$ | $T^2 \propto a^{3.01 \pm 0.01}$

Hooke

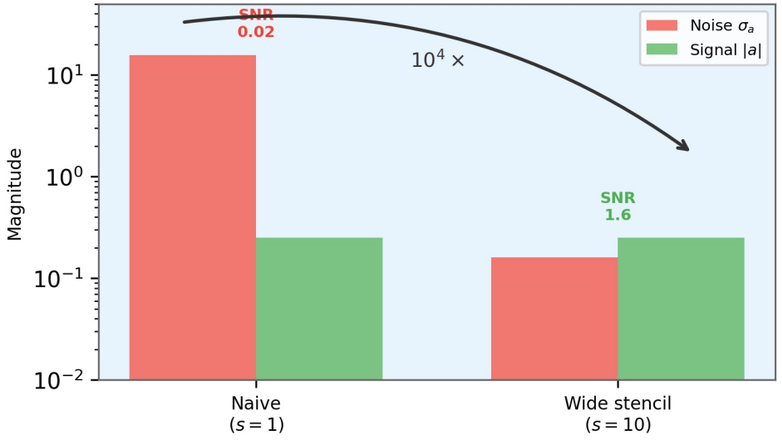
$F = -k r \hat{r}$

$\hat{k} = 0.980 \pm 0.001$

Method comparison

	MAL	SINDy	HNN	LNN
Symbolic?	✓	✓	✗	✗
Validated?	✓	✗	N/A	Failed
# Params	**~20**	~10	17K	17K
Energy	**0.07**	<0.001	0.006	0.013

Wide-stencil noise reduction



MinActionNet Architecture

Basis: $\{r^{-2}, r^{-1}, r, 1, r^{-3}\}$ $\longrightarrow$ softmax gates A_i

Symplectic Velocity-Verlet integrator | SO(2) symmetric | ~ 20 params

Kepler 10/10

biased init or Noetherian selection

Hooke 9/10

direct selection (no post-hoc needed)

Minimum-Action Learning | ~20 parameters | 0.07 kWh | Symbolic + Noether-validated